

Paper: 1MA1/2H				
Question	Answer	Mark	Mark scheme	Additional guidance

19	$9 < m < 11$ $-11 < m < -9$	M1	for a correct method to begin rearranging to solve for m^2 eg $88 < m^2 + 7$ or $m^2 + 7 < 128$ or $81 < m^2 < 121$	It is insufficient to just multiply all three elements by 4; some rearrangement must occur such as showing as two separate inequalities or isolating m^2
		M1	for a complete method to $m^2 = 81$ or $m^2 = 121$ or better	Accept an inequality used in place of “=”. m^2 must be isolated at this stage.
		M1	for a set of critical values: at least two out of $9, 11, -9, -11$	Do not award if other values are also given eg 10
		M1	for selecting a correct inequality for one set of critical values eg $9 < m$ and $m < -9$ or $m < 11$ and $-11 < m$ or $9 < m$ and $m < 11$ or a set of inequalities with some error eg $9 ? m ? 11$ and $-11 ? m ? -9$ where ? is an incorrect inequality symbol like $9 < m \leq 11$ or $9 \geq m \geq 11$ or answer given as $\pm 9 < m < \pm 11$	Could be shown as $9 < m < 11$ or $-11 < m < -9$ or $-11 < m < 11$
		A1	$9 < m < 11$ and $-11 < m < -9$ given as boundaries of m	Accept with an “and” or an “or” or neither